camera

File: camera.kicad_sch

eth

File: eth.kicad_sch

flash

File: flash.kicad_sch

io

File: io.kicad_sch

power

File: power.kicad_sch

sd

File: sd.kicad_sch

soc

File: soc.kicad_sch

ui

File: ui.kicad_sch

usb

File: usb.kicad_sch

LP-1106: Linux SBC on Rockchip RV1106G3 (Cortex-A7, 1 TOPS NPU, 256MB DDR3L in package). USB-C power+OTG, 10/100 Ethernet, SPI NAND boot, microSD, 2-lane MIPI CSI, debug UART, 2x10 GPIO.
- 5V from USB-C (sink, 5.1k Rd), reverse-current Schottky to VSYS
- Rails per RV1106 DS Table 3-2: VDD_0V9 0.81-0.95V, VCC_DDR 1.283-1.45V (DDR3L), VCC_1V8 1.62-1.98V, VCC_3V3 3.0-3.465V
- Sequencing like Luckfox reference: 0.9V first, then DDR+1.8V (RC delay), then 3.3V
- 4-layer JLC 7628 stackup, single-sided assembly, In1=GND plane
- Boot from SPI NAND (FSPI), microSD on SDMMC0, console UART2_M1 115200

PWR_FLAG

ETH_AVDD3V3

rail ETH_AVDD3V3

PWR_FLAG

ETH_CHS

rail ETH_CHS

PWR_FLAG

GND

rail GND

PWR_FLAG

VCC_1V8

rail VCC_1V8

PWR_FLAG

VCC_1V8_CODEC

rail VCC_1V8_CODEC

PWR_FLAG

VCC_1V8_MIPI

rail VCC_1V8_MIPI

PWR_FLAG

VCC_1V8_PLL

rail VCC_1V8_PLL

PWR_FLAG

VCC_3V3

rail VCC_3V3

PWR_FLAG

VCC_DDR

rail VCC_DDR

PWR_FLAG

VCC_RTC

rail VCC_RTC

PWR_FLAG

VDD_0V9

rail VDD_0V9

PWR_FLAG

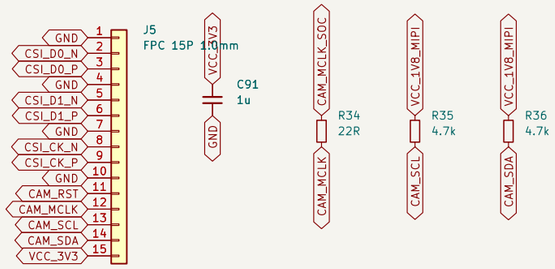
VSYS

rail VSYS

Sheet: /
File: rv1106_sbc.kicad_sch

**Title: LP-1106 RV1106 SBC**

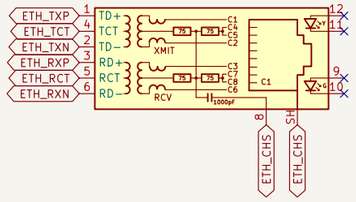
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|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 1/10 |

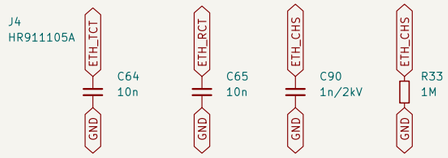



Sheet: /camera/
File: camera.kicad_sch

**Title: LP-1106 RV1106 SBC - camera**

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 2/10 |

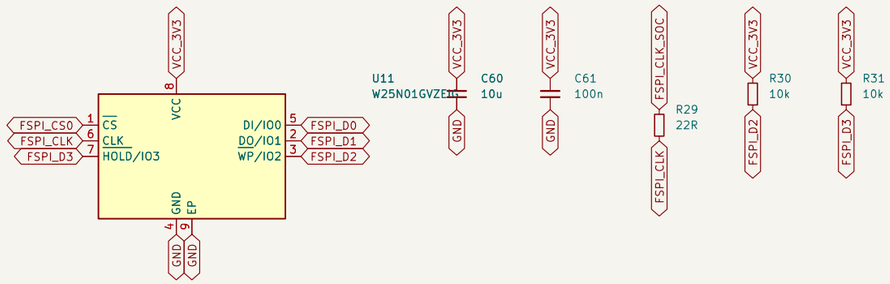



Sheet: /eth/
File: eth.kicad_sch

**Title: LP-1106 RV1106 SBC - eth**

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 3/10 |



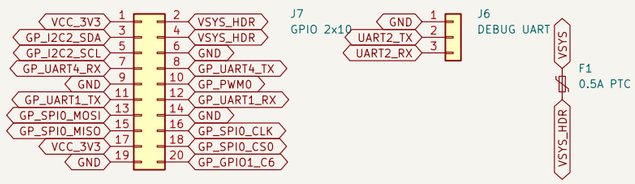




Sheet: /flash/
File: flash.kicad_sch

**Title: LP-1106 RV1106 SBC - flash**

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 4/10 |

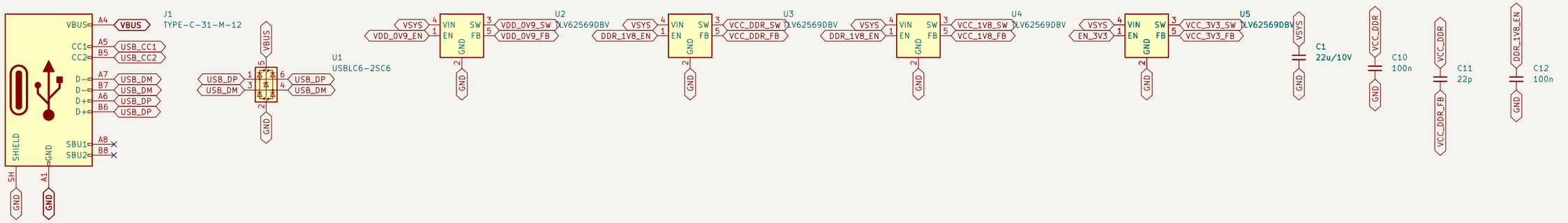


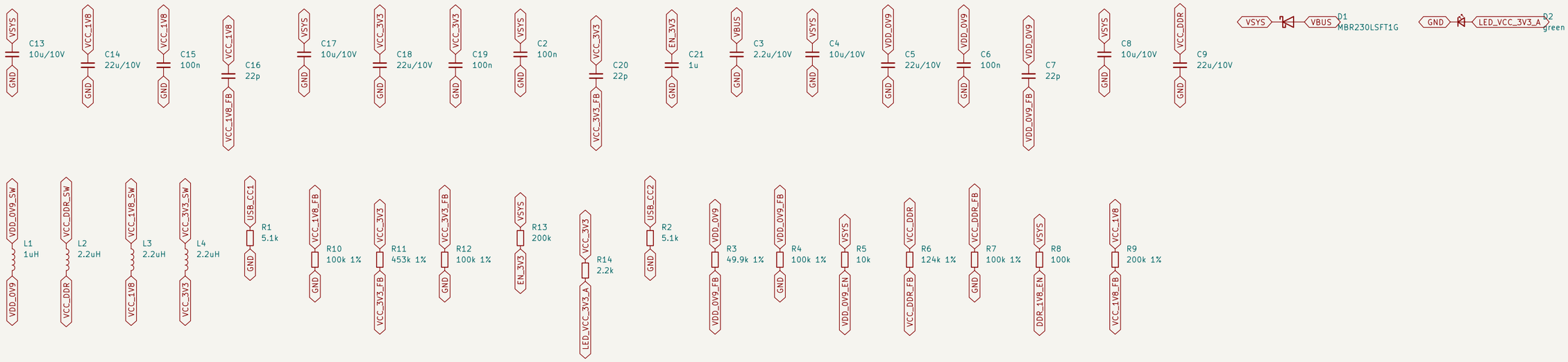




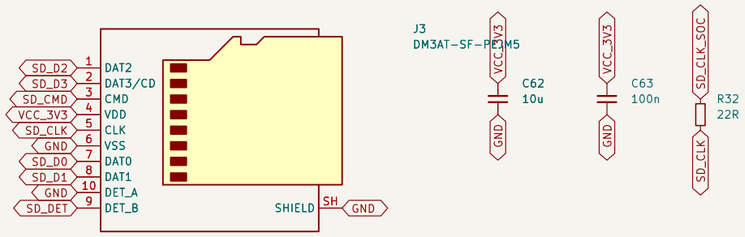



| | | |
|---|---|---|
| Sheet: /io/ | | |
| File: io.kicad_sch | | |
| **Title: LP-1106 RV1106 SBC - io** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.8 | | Id: 5/10 |




| | | |
|---|---|---|
| Sheet: /power/ | | |
| File: power.kicad_sch | | |
| **Title: LP-1106 RV1106 SBC - power** | | |
| Size: A3 | Date: | **Rev: A** |
| KiCad E.D.A. 10.0.8 | | Id: 6/10 |









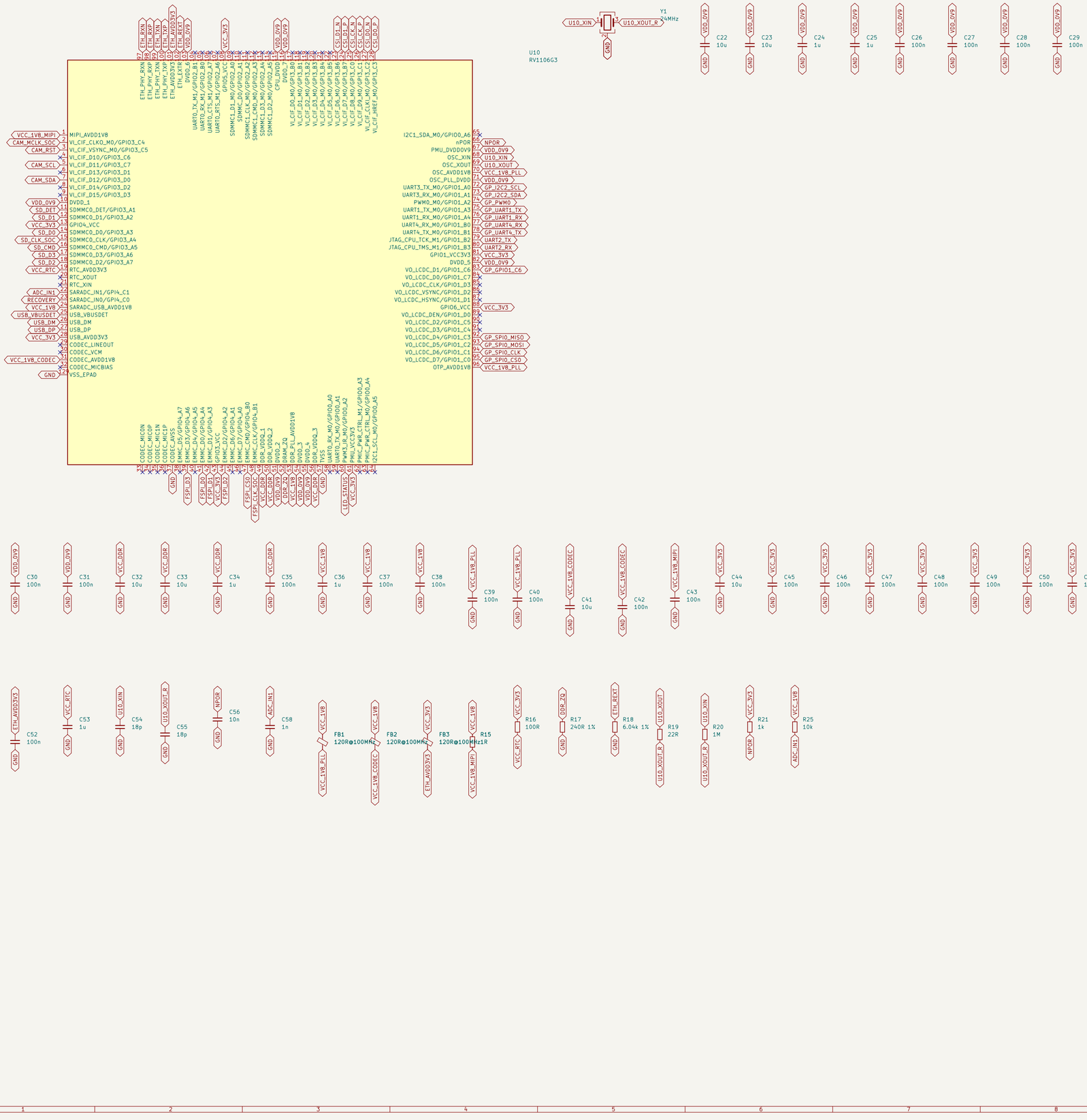




Sheet: /sd/
File: sd.kicad_sch

**Title: LP-1106 RV1106 SBC - sd**

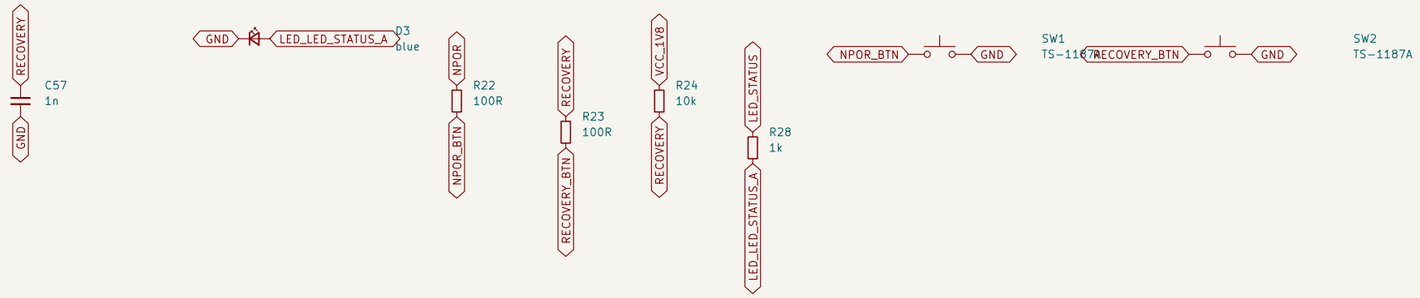
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|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 7/10 |




Sheet: /soc/
File: soc.kicad_sch

**Title: LP-1106 RV1106 SBC - soc**

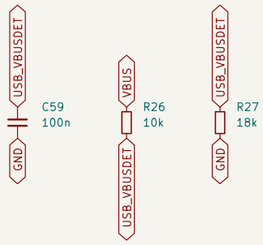
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|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 8/10 |




Sheet: /ui/
File: ui.kicad_sch

**Title: LP-1106 RV1106 SBC - ui**

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 9/10 |




Sheet: /usb/
File: usb.kicad_sch

**Title: LP-1106 RV1106 SBC - usb**

| Size: A3 | Date: | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 10/10 |